

Situation 1

An investigator is interested in assessing whether there is an association between nitrate (NO) levels in healthy individuals and patients with hypertension.

A random sample of 4 healthy individuals and their NO levels (in nM) were:

0.97, 0.95, 0.92, 0.89

whereas, for the sample of 5 patients diagnosed with hypertension, yield the following NO values

0.60, 0.81, 0.77, 0.67, 0.74

The research question is whether there is statistical evidence that the average NO concentration in healthy individuals is higher than that in hypertensive patients. Set the hypothesis test to answer the question with a significance level of $\alpha=0.05$.

We remind you that before conducting the comparison of means, it is necessary to verify the assumption of homogeneity of variances.

First, we are going to carry out the test of two-variances (F-test) with the following hypotheses:

$$\begin{cases} H_0 : \sigma_1^2 = \sigma_2^2 \\ H_1 : \sigma_1^2 \neq \sigma_2^2 \end{cases}$$

where σ_1^2 is the variance of the NO level among healthy individuals and σ_2^2 is the variance of the NO level among hypertensive patients.

1. Which is the alternative hypothesis of the test of two variances

Solution:

The population variance of the NO level for healthy individuals is different from the population variance of the NO level for hypertensive individuals.

2. Calculate the sample standard deviation of NO level for the healthy individuals.

Solution:

Healthy:	$n_1 = 4$	$\bar{x}_1 = 0.9325$	$s_1 = 0.035$	$s_1^2 = 0.0012$
Hypertensive:	$n_2 = 5$	$\bar{x}_2 = 0.718$	$s_2 = 0.0835$	$s_2^2 = 0.007$

3. Calculate the test statistic for the F-test, considering that the healthy group goes to the numerator.

Solution:

The test statistic corresponding to the variance test is

$$\frac{s_1^2}{s_2^2} = \frac{0.035^2}{0.0835^2} = \frac{0.0012}{0.007} = 0.1757$$

which follows a Snedecor F distribution with degrees of freedom $n_1 - 1 = 3$ in the numerator and $n_2 - 1 = 4$ in the denominator.

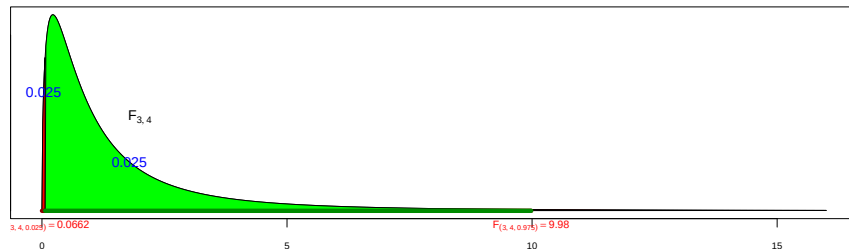
4. Calculate the acceptance interval with $\alpha=0.05$ of the F-test.

Solution: The lower endpoint of the interval is

$$F_{n_1-1, n_2-1; \alpha/2} = F_{3,4;0.025} = \frac{1}{F_{4,3;0.975}} = \frac{1}{15.1} = 0.0662$$

and the upper endpoint is

$$F_{n_1-1, n_2-1; 1-\alpha/2} = F_{3,4;0.975} = 9.98$$



5. Which is the most appropriate conclusion?

Solution: Since the test statistic = 0.1757 belongs to the interval $[0.0662, 9.98]$. We failed to reject the null hypothesis. There are no significant differences between the population variances. That is, it can be assumed that the population variances of both populations are equal.

Finally, we are going to set the following test of two-means or test of independent samples according to our initial research question:

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_1 : \mu_1 > \mu_2 \end{cases}$$

where μ_1 is the population mean of NO level among healthy individuals and μ_2 is the population mean of NO level for hypertensive patients.

6. Which is the alternative hypothesis of the test of two means?

Solution: The population mean of the the NO level for healthy individuals is higher than the population mean of the NO level for hypertensive patients.

7. Calculate the pooled variance.

Solution: We calculate the pooled variance

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(4 - 1)0.035^2 + (5 - 1)0.0835^2}{4 + 5 - 2} = 0.0045$$

8. Calculate the corresponding test statistic.

Solution: We calculate the pooled standard deviation

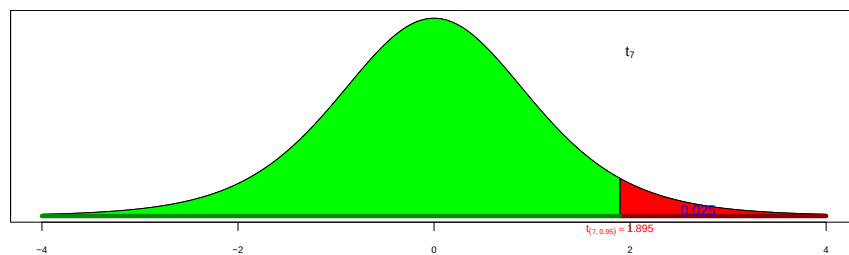
$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(4 - 1)0.035^2 + (5 - 1)0.0835^2}{4 + 5 - 2}} = 0.0671$$

and the test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.9325 - 0.718}{0.0671\sqrt{\frac{1}{4} + \frac{1}{5}}} = 4.7654$$

9. Calculate the acceptance interval

Solution: The significance level is $\alpha=0.05$ and it is one-sided rejecting on the right. Therefore, we have to calculate the percentile at 5% of the Student's t distribution with $n_1 + n_2 - 2 = 7$ degrees of freedom, $t_{(7,0.95)} = 1.895$. The acceptance interval is $(-\infty, 1.895]$



10. Which is the most appropriate conclusion?

Solution:

The test statistic is $t=4.7654$ does not belong to the interval $(-\infty, 1.895]$. Consequently, the null hypothesis is rejected. We have collected enough evidence that the average NO level for healthy individuals is higher than the average NO level for hypertensive patients.

Situation 2

A treatment with a calcium supplement is expected to increase salivary pH level. To test this hypothesis, a research group selected 5 patients and measured their pH values before and after the treatment with the following pH values:

Is the treatment effective to increase salivary pH? Conduct the following hypothesis test to answer the question with a significance level of $\alpha = 0.05$:

$$\begin{cases} H_0 : \mu_{dif} = 0 \\ H_1 : \mu_{dif} > 0 \end{cases}$$

baseline	after treatment
7.50	7.67
7.32	7.36
6.83	6.93
7.73	8.08
7.17	7.06

11. What does μ_{dif} stand for?

Solution: The population mean of the pH level after the treatment minus population mean of pH level at the baseline.

This hypothesis testing can be set on the following ways:

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_1 : \mu_1 < \mu_2 \end{cases}$$

$$\begin{cases} H_0 : \mu_2 - \mu_1 = 0 \\ H_1 : \mu_2 - \mu_1 > 0 \end{cases}$$

where μ_1 is the population mean of the pH level at baseline and μ_2 is the population mean of the pH level after treatment.

12. Calculate the test statistic

Solution: We calculate the differences of the two variables:

Degree of inflammation		
after 24h	before	Difference
7.67	7.50	0.17
7.36	7.32	0.04
6.93	6.83	0.10
8.08	7.73	0.35
7.06	7.17	-0.11

Data: $\bar{x}=0.11$, $s=0.1693$, $n=5$.

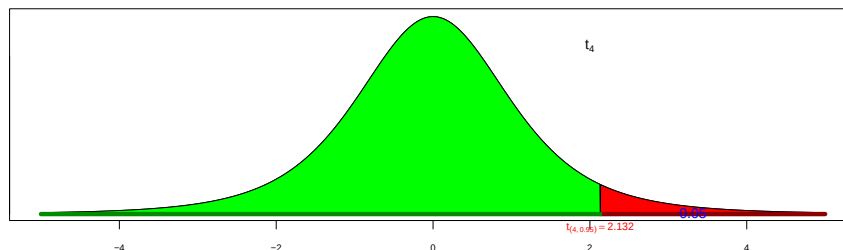
The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{0.11 - 0}{0.1693/\sqrt{5}} = 1.4528$$

which follows a Student's t distribution with $n - 1 = 4$ degrees of freedom

13. Calculate the corresponding acceptance interval

Solution: With the significance level $\alpha=0.05$ and a one-sided hypothesis testing, we calculate the percentile at 95% of the Student's t distribution with 4 degrees of freedom, $t_{(4,0.95)} = 2.132$



The acceptance interval is $(-\infty, 2.132]$

14. Which is the most appropriate conclusion?

Solution:

The test statistic 1.4528 belongs to the interval $(-\infty, 2.132]$. Hence, the null hypothesis is failed to reject. The mean pH after the treatment is significantly higher than the pH level at the baseline.

Situation 3

Note: For this exercise, the expected values are recommended to be rounded off to two decimal digits. All the other calculations use 4 digits as usual.

A research group aims to determine whether the type of irrigant used in endodontic treatment is associated with the occurrence of adverse events. To test this hypothesis, 946 patients were randomly assigned to one of the three irrigant groups: Sodium Hypochlorite (NaOCl), Chlorhexidine (CHX), and Ethylenediaminetetraacetic Acid (EDTA).

Five years after treatment, the researchers recorded whether each patient experienced an adverse event. The results of the study are summarized in the following table.

	NaOCl	CHX	EDTA
Adverse events	13	10	18
No problem	202	401	302

Use a significance level $\alpha=0.05$.

15. Describe the null hypothesis.

Solution: The null hypothesis is

There is NOT association between the employed irrigant and experiencing an adverse event. Adverse events does NOT depend on irrigant.

and the alternative hypothesis is that there exists an association between the variables

There is association between the employed irrigant and experiencing an adverse event. The chances of adverse events depends on the irrigant used.

16. The row total of individuals with absence of adverse events is ...

Solution: The observation table with row and column totals

	NaOCl	CHX	EDTA	Row total
Adverse events	13	10	18	41
No problem	202	401	302	905
Column total	215	411	320	946

The value is 905.

17. The column total of individuals using NaOCl is ...

Solution: The value is 215

18. The expected value of the occurrence of an adverse event and using NaOCl is ...

Solution: The expected values table is

	NaOCl	CHX	EDTA
Adverse event	$\frac{41 \cdot 215}{946} = 9.32$	$\frac{41 \cdot 411}{946} = 17.81$	$\frac{41 \cdot 320}{946} = 13.87$
No problem	$\frac{905 \cdot 215}{946} = 205.68$	$\frac{905 \cdot 411}{946} = 393.19$	$\frac{905 \cdot 320}{946} = 306.13$

The value is 9.32

19. The expected value of the occurrence of an adverse event and using EDTA is ...

Solution: The value is 13.87

20. Calculate the test statistic

Solution:

$$x^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

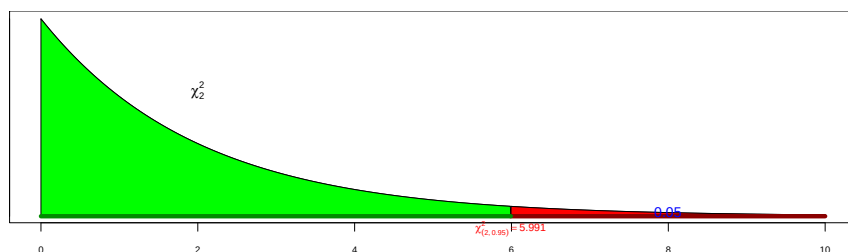
$$= \frac{(13 - 9.32)^2}{9.32} + \frac{(10 - 17.81)^2}{17.81} + \frac{(18 - 13.87)^2}{13.87} + \frac{(202 - 205.68)^2}{205.68} + \frac{(401 - 393.19)^2}{393.19} + \frac{(302 - 306.13)^2}{306.13}$$

$$1.4530 + 3.4248 + 1.2298 + 0.0658 + 0.1551 + 0.0557 = 6.3842$$

which follows a chi-squared distribution with $(2 - 1) \cdot (3 - 1) = 2$ degrees of freedom.

21. Calculate the acceptance interval with significance level $\alpha=0.05$

Solution: The acceptance interval is $[0, \chi_{2,0.95}^2]$. On the table que check that the values is 5.991.



22. Write an appropriate conclusion of the test of independence with significance level $\alpha=0.05$.

Solution:

Since the test statistic $x^2=6.3842$ is not in the interval $[0, 5.991]$, hence we have enough evidence of association between the occurrences of adverse events and the type of irrigant used.

Situation 4

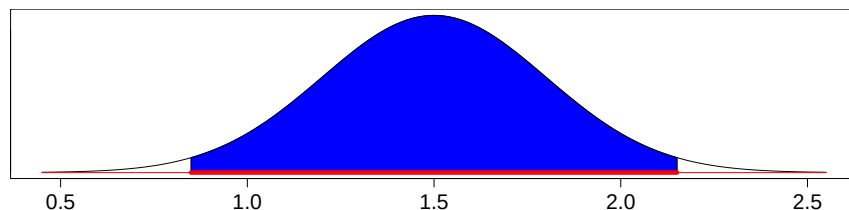
We consider the salivary calcium level of a healthy adult individuals to a variable normally distributed with mean $\mu=1.5$ and standard deviation $\sigma=0.3$.

23. Calculate a mean-centred interval containing the 97% of the salivary calcium levels.

Solution: The central 97% of the values of IQ are between the 1.5 and the 98.5 percentiles. First, we calculate these values of the standard normal distribution, $z_{0.015} = -2.17$ and $z_{0.985} = 2.17$. Second, we calculate the corresponding values of the normal distribution with $\mu = 1.5$ and $\sigma = 0.3$.

$$x_{0.015} = \mu - \sigma z_{0.985} = 1.5 - 0.3 \cdot 2.17 = 0.849$$

$$x_{0.985} = \mu + \sigma z_{0.985} = 1.5 + 0.3 \cdot 2.17 = 2.151$$



24. What is the proportion of people with a salivary calcium level between 1.09 and 1.76?

Solution: We calculate the z-score of $x = 1.09$ y $x = 1.76$,

$$\begin{aligned} x = 1.09 & \quad z = \frac{x - \mu}{\sigma} = \frac{1.09 - 1.5}{0.3} = -1.37 \\ x = 1.76 & \quad z = \frac{x - \mu}{\sigma} = \frac{1.76 - 1.5}{0.3} = 0.87 \end{aligned}$$

Hence, we calculate the probability of Z , the standard normal distribution, between -1.37 and 0.87.

$$\begin{aligned} P(1.09 < X < 1.76) &= P(-1.37 < Z < 0.87) = P(Z < 0.87) - P(Z < -1.37) = \\ &= P(Z < 0.87) - P(Z > 1.37) = 0.8078 - (1 - 0.9147) = 0.7225 \end{aligned}$$

$$P(1.09 < pH < 1.76) =$$

